Income (₹)	0-100	100-200	200-300	300-400	400-500	500-600	600-700
No.of persons	5	7	12	18	16	10	5

Solution:

Income (₹)	No.of persons (f)	
0-100	5	
100-200	7	
200-300	12	f_0
300-400	18	f_1
400-500	16	f_2
500-600	10	
600-700	5	

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times C$$

The highest frequency is 18, the modal class is 300-400

Here, $l = 300$, $f_0 = 12$, $f_1 = 18$, $f_2 = 16$,

$$\text{Mode} = 300 + \frac{18 - 12}{2 \times 18 - 12 - 16} \times 100$$

$$= 300 + \frac{6}{36 - 28} \times 100$$

$$= 300 + \frac{6}{8} \times 100$$

$$= 300 + \frac{600}{8} = 300 + 75 = 375$$

The modal income of the families is 375.

Determination of Modal class:

For a frequency distribution modal class corresponds to the class with maximum frequency. But in any one of the following cases that is not easily possible.

(i) If the maximum frequency is repeated.

(ii) If the maximum frequency occurs in the beginning or at the end of the distribution

(iii) If there are irregularities in the distribution, the modal class is determined by the method of grouping.

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